# 📊 Sales Performance & Profit Analysis Using Power BI

## 🎯 Project Objective:

The objective of this project is to analyze sales performance across different categories ,sub-categories, cities,and states using Power BI. The workflow includes importing multiple datasets, performing ETL operations, creating relationships between tables, developing calculated fields using DAX, and designing an interactive dashboard for decision-making. The dashboard helps stakeholders monitor sales, profit, targets, quantity sold and order distribution to identify business trends and make data-driven decisions.

## 🗃 Dataset Description:

This dataset contains sales, customer, product, and order-related information used to perform data transformation, data modeling, and business analysis in Power BI. The purpose of the dataset is to analyze business performance, identify sales trends, evaluate profitability, and generate actionable insights through interactive dashboards. And multiple related tables were integrated in power BI to create an interactive dashboard using data modeling and DAX calculations.

The dataset includes multiple related tables such as:

- ✓ **Orders Table** – Contains order details including Order ID, Order Date, Customer ID, Product ID, Quantity, Sales, and Discount.
- ✓ **Customers Table** – Stores customer information such as Customer ID, Customer Name, Segment, Region, and other demographic details.
- ✓ **Products Table** – Includes product-related details like Product ID, Product Name, Category, and Sub-Category.
- ✓ **Sales/Profit Table** – Contains revenue, cost, and profit-related metrics used for financial analysis.
- ✓ **Targets Table** – Stores sales targets for performance comparison and KPI tracking.
- ✓ **List of Orders** - Contains order details and customer location information.
- ✓ **Order Details** - Includes product sales transaction s and quantities.
- ✓ **Sales Target** - Stores target values for performance comparison.
- ✓ **Order Profit** - Contains profit values for profitability analysis.

### Key Objectives of the Dataset

- ✧ Clean and transform raw business data using ETL processes.

- ✧ Build relationships between multiple tables through data modeling.
- ✧ Create calculated columns and DAX measures.
- ✧ Analyze sales performance and profitability.
- ✧ Develop interactive dashboards and reports for business decision-making.
- ✧ Total Sales
- ✧ Total Profit
- ✧ Total Order s
- ✧ Profit Margin (%)
- ✧ Target Achievement (%)
- ✧ Monthly Sales Trend
- ✧ Category & Sub-category performance
- ✧ City/State-wise Sales Distribution

## Tools Used

- **Power BI Desktop** – Data Modeling & Dashboard Development
- **Power Query Editor** – Data Cleaning & Transformation
- **DAX (Data Analysis Expressions)** – Calculations and KPI creation

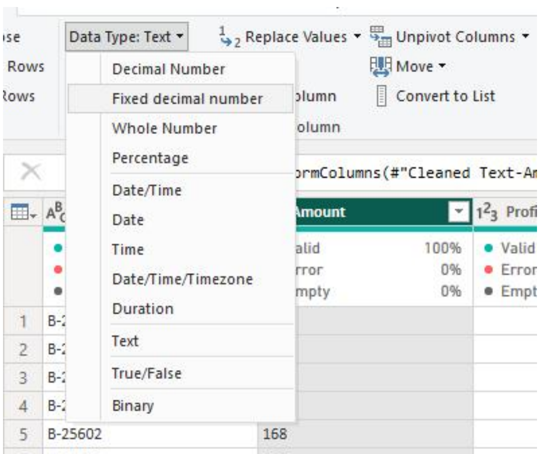
## 📊 Column Description

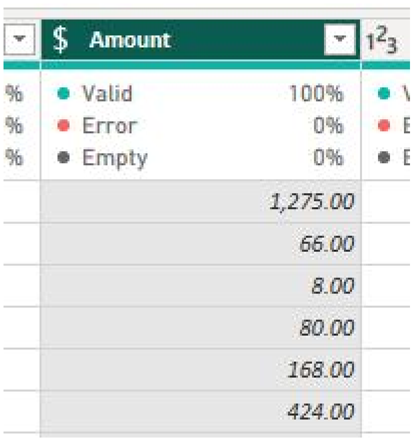
| Column Name | Description |
|---|---|
| Order ID | Unique identifier for each order |
| Order Date | Date when the order was placed |
| Customer ID | Unique identifier for each customer |
| Customer Name | Name of the customer |
| Product ID | Unique identifier for each product |
| Product Name | Name of the purchased product |
| Category | Product category (Electronics, Clothing, etc.) |

| | |
|---|---|
| Quantity | Number of units purchased |
| Unit Price | Price per unit of the product |
| Discount | Discount applied to the order |
| Revenue | Total sales amount generated |
| Cost | Total cost of products sold |
| Profit | Revenue minus cost |
| Region | Geographic sales region |
| Sales Target | Expected sales value |
| State | State where the order was placed |
| City | City where the order was placed |
| Sub-Category | Product sub-category |
| Target | Sales target assigned |
| Month | Sales month for trend analysis |

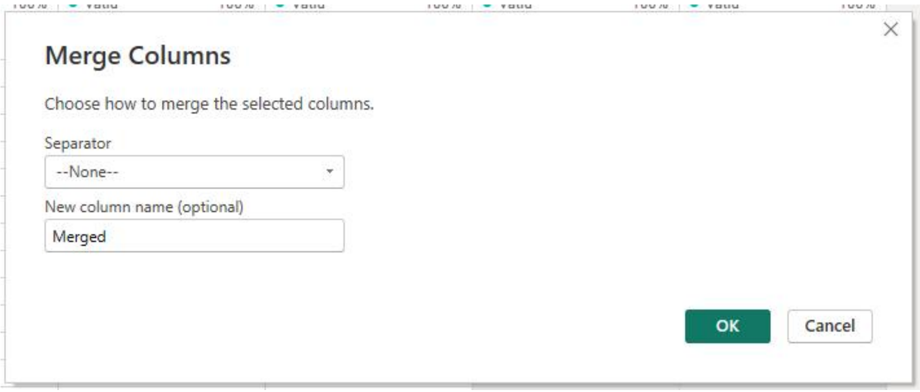
## Data Cleaning & Transformation

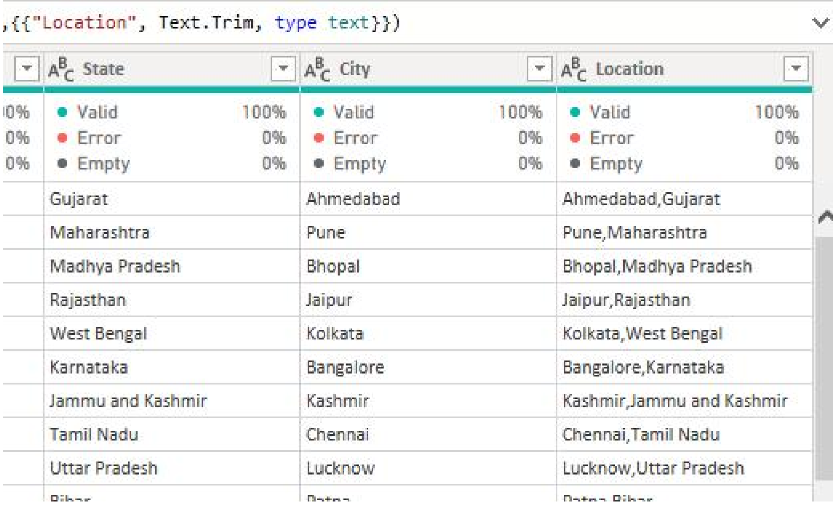
- Imported raw datasets into Power BI and verified data types.
- Removed duplicate records to maintain data accuracy.
- Handled missing values using replacement rules and transformations.
- Standardized text fields (customer names, categories, IDs).
- Corrected data types for columns(Text, Number,Data, Currency)
- Created calculated columns and measures using DAX where required.
- Established relationships between tables for efficient data modeling.
- Applied transformations using Power Query Editor:
    - **Change Data Types**



- ## Split/Merge Columns

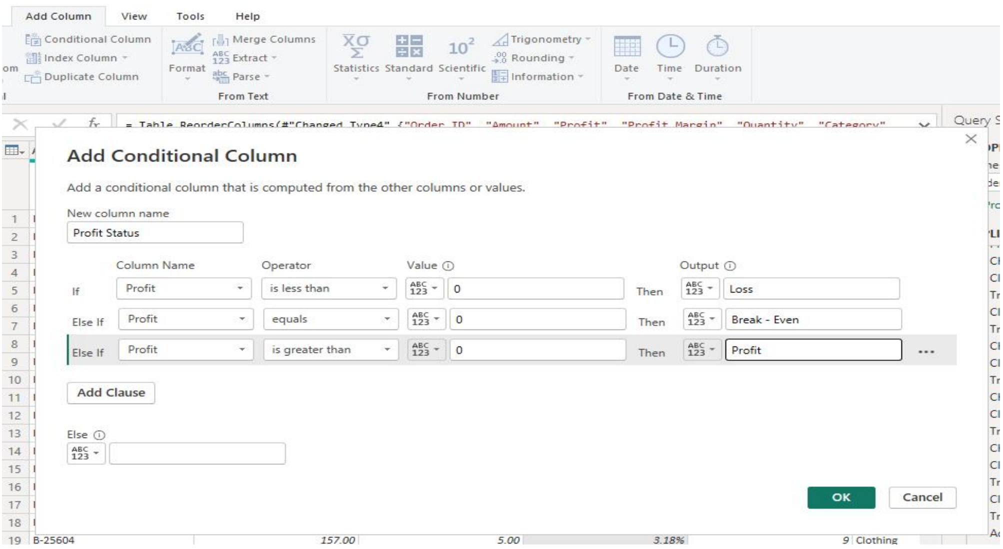


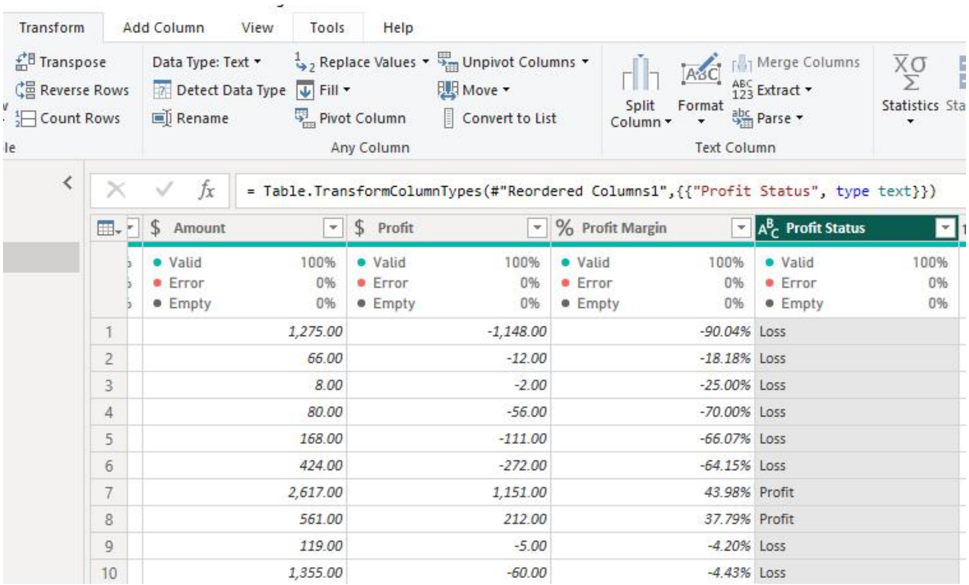




- ## Conditional Columns

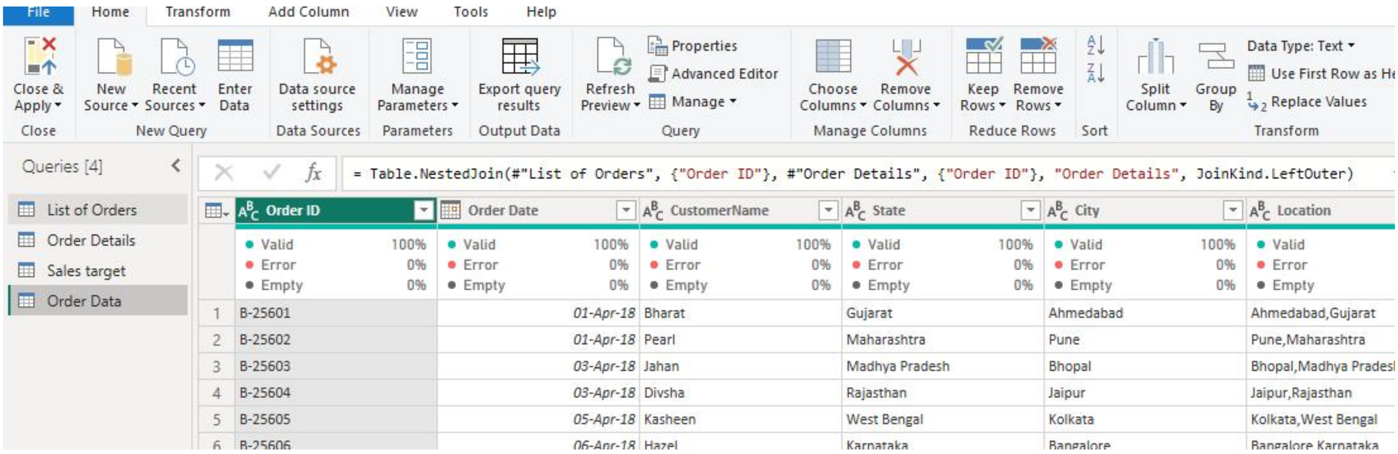
## Add a conditional column named Profit Status based on:

- **Profit < 0 → *Loss***
- **Profit = 0 → *Break-Even***
- **Profit > 0 → *Profit***






- **Data Validation**




# Data Transformation:

- Merged related tables using **Order ID** and category fields.
- Created relationships between:
  - **List of Orders → Order Details**
  - **Order Details → Sales Target**
  - **Order Details → Order Profit**
- Created calculated measures using **DAX**, including:
  - **Total Sales = SUM(Amount)**
  - **Total Profit = SUM(Profit)**
  - **Profit Margin = DIVIDE(Total Profit, Total Sales, 0)**
  - **Target Achievement %**
- Generated date-based fields for **monthly sales trend analysis**.
- Created KPIs to compare **Actual Sales vs Target Sales**.

- Applied aggregations and filters for interactive reporting.
- Built hierarchies for **Category → Sub-Category** and **State → City** analysis.

# 🧠 Visualization and Insights

**(Attach Screenshot of Visualizations)**

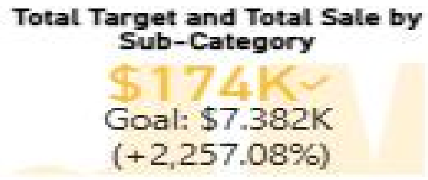
Include screenshots of:

- **KPI Cards**

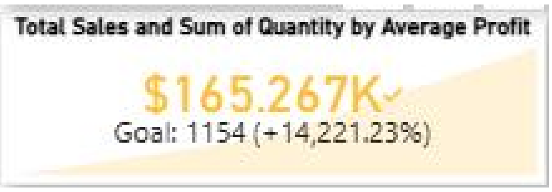
1. Monthly Sales and Total Sales by Profit


2. Minimum Target and Count of Order ID by Target


3. Total Target and Total Sales by Sub-Category




4. Total Sales and Sum of Quantity by Average



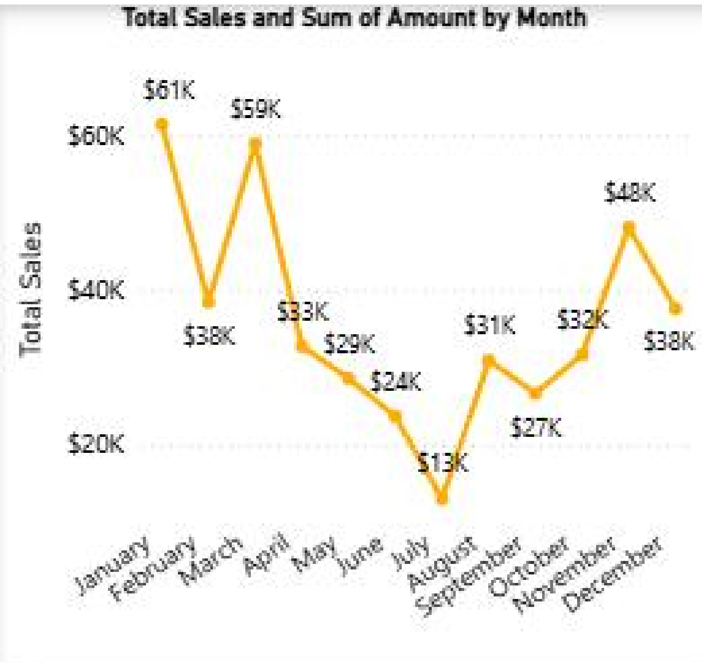

- **Matrix / Table Visual**

1. Quantity, Clothing, Electronics, Furniture, Total

| Month | Clothing | Electronics | Furniture |
|---|---|---|---|
| December | $139,054 | $165,267 | $127,181 |
| November | $139,054 | $165,267 | $127,181 |
| October | $139,054 | $165,267 | $127,181 |
| September | $139,054 | $165,267 | $127,181 |
| August | $139,054 | $165,267 | $127,181 |
| July | $139,054 | $165,267 | $127,181 |
| June | $139,054 | $165,267 | $127,181 |
| May | $139,054 | $165,267 | $127,181 |
| April | $139,054 | $165,267 | $127,181 |
| March | $139,054 | $165,267 | $127,181 |
| February | $139,054 | $165,267 | $127,181 |
| January | $139,054 | $165,267 | $127,181 |
| **Total** | $139,054 | $165,267 | $127,181 |

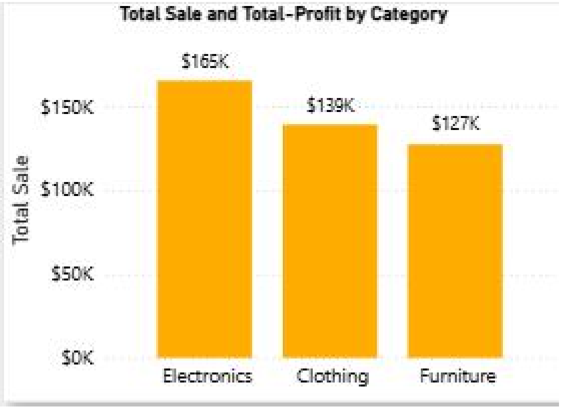
## ● Line Chart

1. Total Sales and Sum of Amount by Month

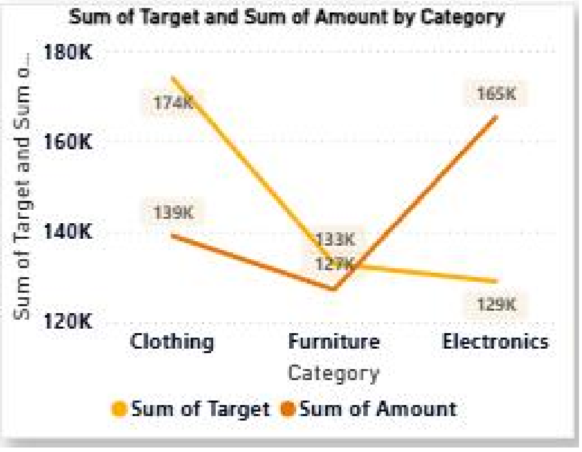



- **Bar Chart**

- **Stacked Column Chart**

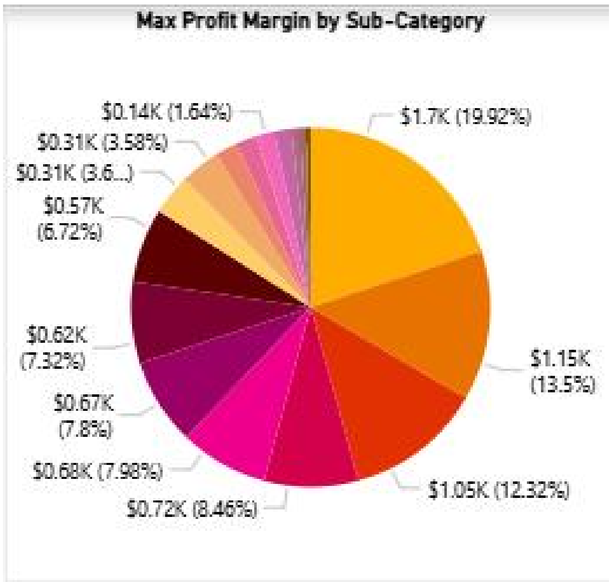



- **Line Chart**

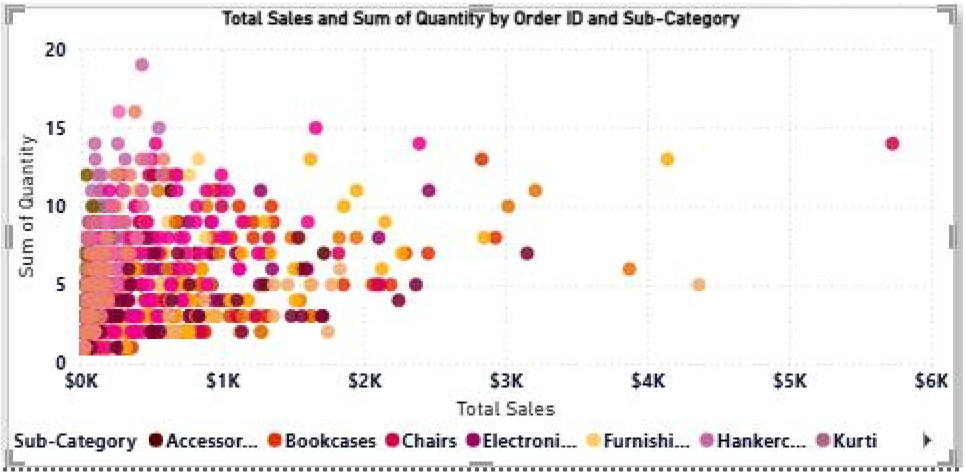


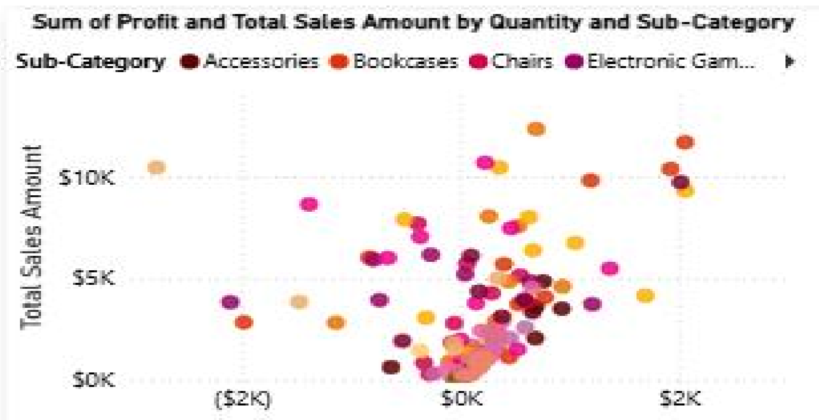

- **Pie/Donut Chart**


1. Max Profit Margin by Sub-Category




- **Scatter Chart**




- **Gauge Chart**


- **Table**

| Customer Name | Total Sales |
|---|---|
| Ashwin | $11 |
| Dinesh | $12 |
| Hemangi | $13 |
| Rohit | $15 |
| Ananya | $16 |
| Anubhaw | $16 |
| Vipul | $16 |
| Saloni | $17 |
| Ashvini | $22 |
| Sumeet | $24 |
| Total | $431,502 |

| Month | Clothing | Electronics | Furniture |
|---|---|---|---|
| December | $139,054 | $165,267 | $127,181 |
| November | $139,054 | $165,267 | $127,181 |
| October | $139,054 | $165,267 | $127,181 |
| September | $139,054 | $165,267 | $127,181 |
| August | $139,054 | $165,267 | $127,181 |
| July | $139,054 | $165,267 | $127,181 |
| June | $139,054 | $165,267 | $127,181 |
| May | $139,054 | $165,267 | $127,181 |
| April | $139,054 | $165,267 | $127,181 |
| March | $139,054 | $165,267 | $127,181 |
| February | $139,054 | $165,267 | $127,181 |
| January | $139,054 | $165,267 | $127,181 |
| Total | $139,054 | $165,267 | $127,181 |

- **Map**


## Key Insights:

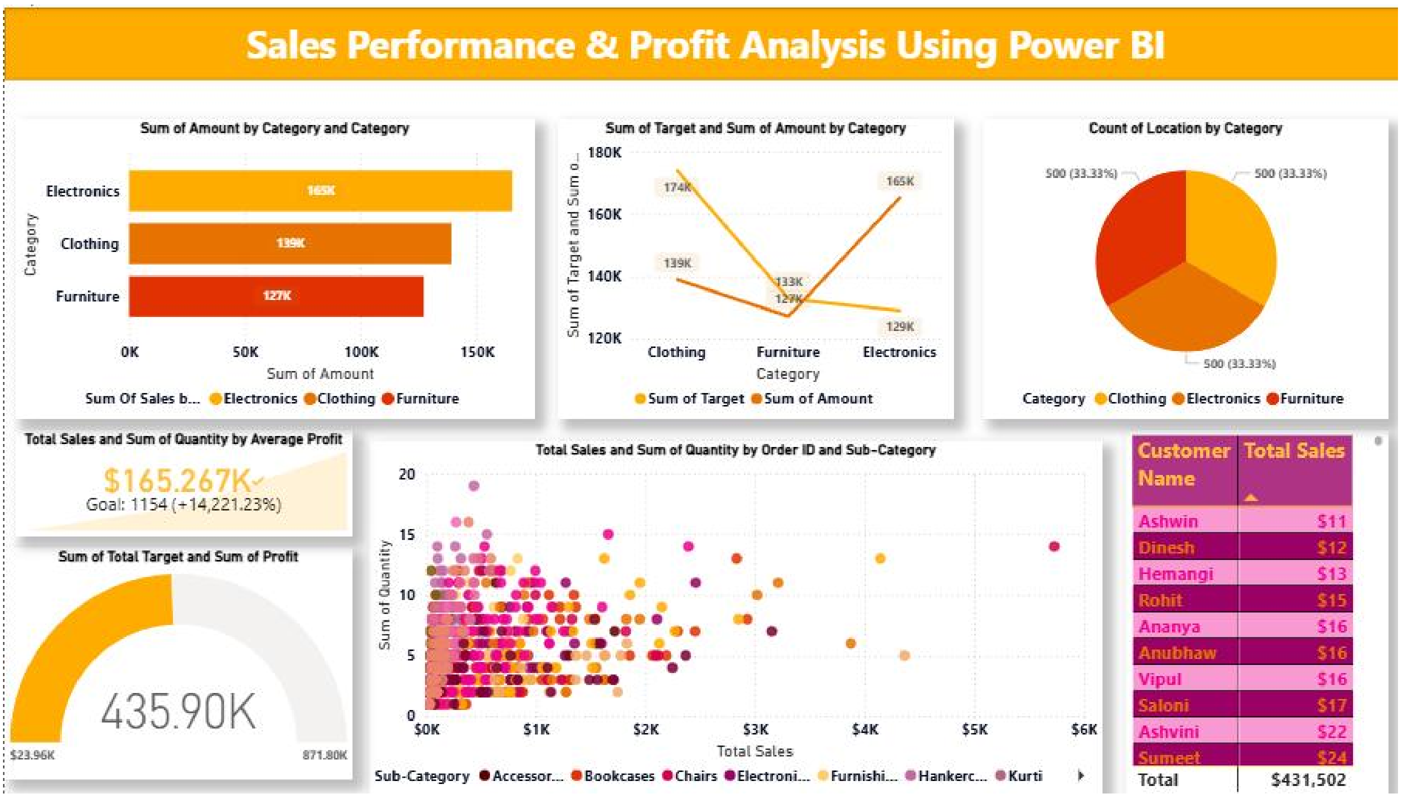
- Highest revenue-generating category identified.
- Top-performing products contributed major sales volume.
- Profit varied across categories and regions.
- Seasonal/monthly trends observed in customer purchasing behavior.

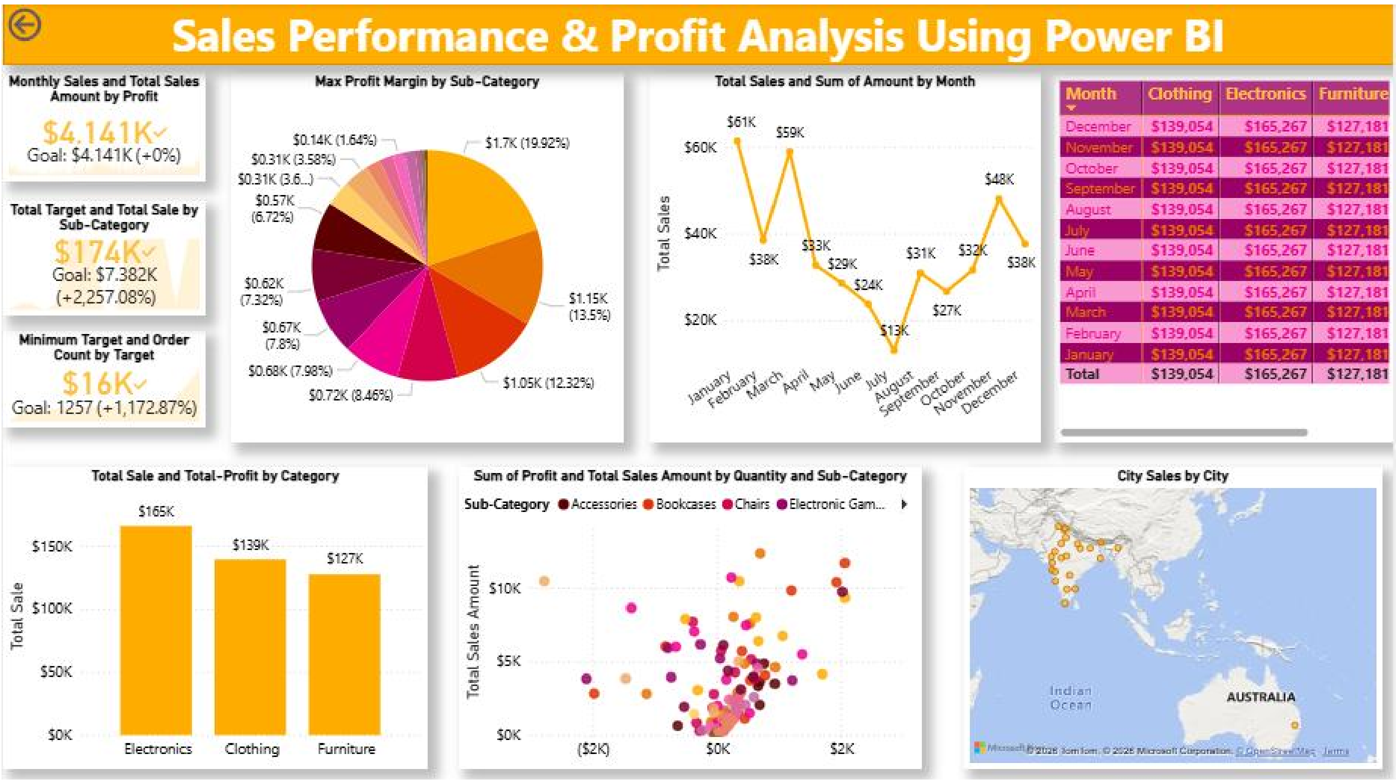
## Report / Dashboard Screenshot

Insert final dashboard screenshot here.

## Dashboard Components:

- KPI Summary
- Sales Performance
- Profit Analysis
- Product Performance
- Interactive Filters






## ⚙ Analysis

### 1. Descriptive Analysis

**Purpose:**

Summarize historical data and show overall performance.

- Total Sales achieved: **$165.267K**
- Total Profit: **435.90K**
- Category-wise sales:
  - Electronics → Highest sales (~156K)
  - Clothing → ~136K
  - Furniture → ~121K
- Customer-wise sales table shows top contributing customers.

**Key Insight:**
Electronics generated the highest revenue compared to other categories.

### 2. Diagnostic Analysis

**Purpose:**

Identify reasons behind performance.

- Compare **Sales vs Target by Category**
  - Electronics exceeded target.
  - Furniture showed lower sales and missed target.

- Scatter chart (**Sales vs Quantity**) shows:
    - Higher quantity orders do not always generate higher sales.
    - Some products generate high sales with fewer quantities.

**Key Insight:**
Category performance differences and product demand influence overall revenue.

### 3. Predictive Analysis

**Purpose:**

Use trends to forecast future outcomes.

- Electronics sales may continue growing if demand remains stable.
- Furniture could continue underperforming unless strategy changes.
- High-performing customers may contribute more future revenue.

**Prediction:**
If current sales trends continue, Electronics is expected to remain the leading category.

### 4. Prescriptive Analysis

**Purpose:**

Recommend actions for improvement.

**Recommendations:**

- Increase inventory and promotions for Electronics.
- Introduce discounts or campaigns for Furniture.
- Focus customer retention programs on top buyers.
- Optimize product mix based on quantity vs sales relationship.

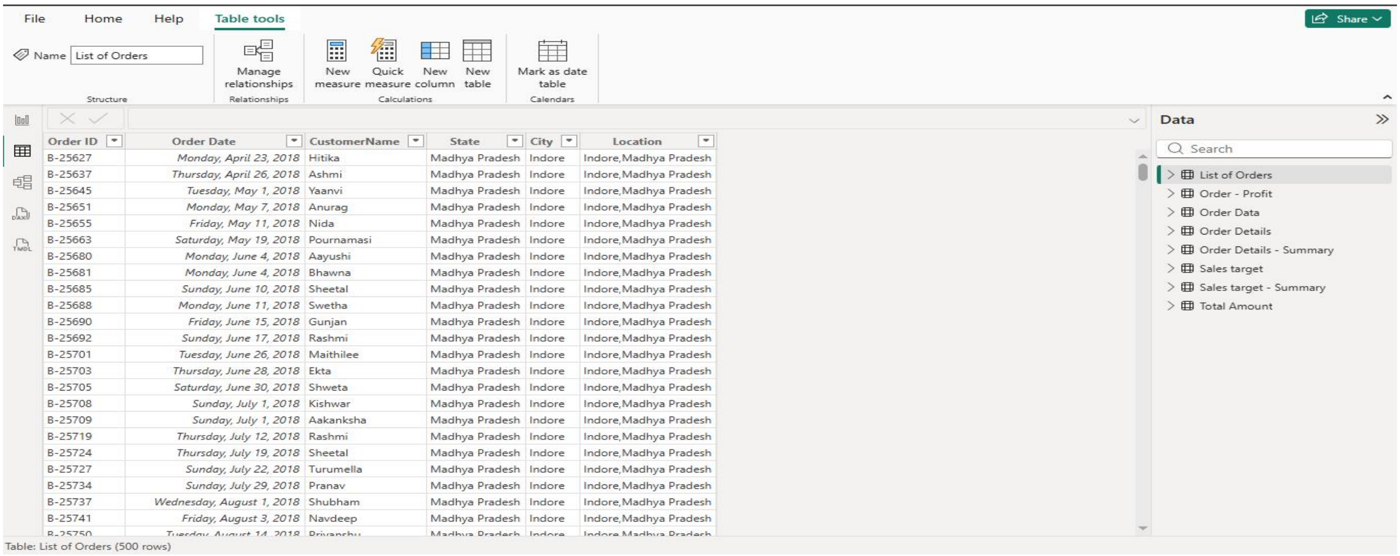
**Action Plan:**

- Maintain high-performing categories.
- Improve low-performing segments.
- Use customer insights for targeted marketing.

## 🎯 Conclusion

This project transformed raw data into meaningful business insights using Power BI. Through data cleaning, modeling, visualization, and analytical techniques, trends and performance drivers were identified. The dashboard supports data-driven decision-making by helping stakeholders monitor KPIs, understand customer behavior, and improve business performance

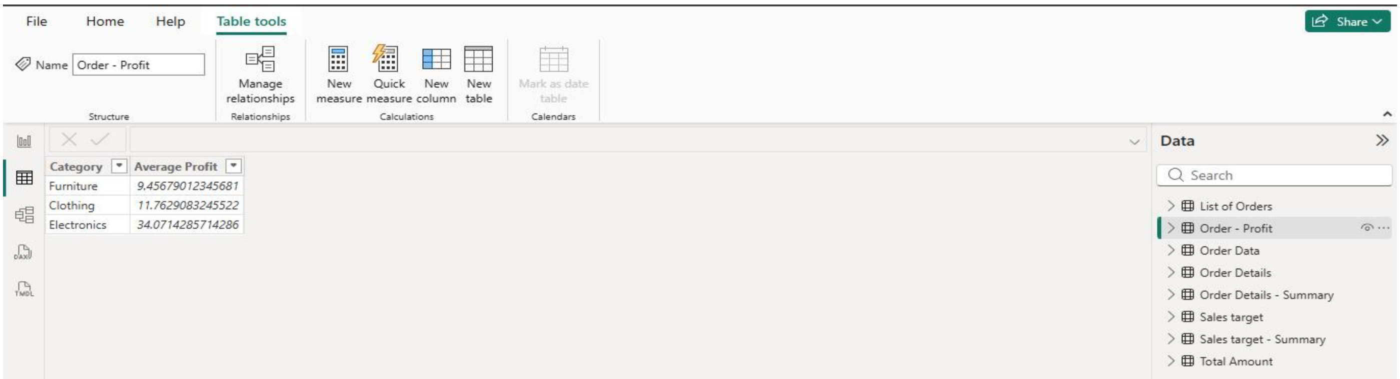
## 1. LIST OF ORDER



| Order ID | Order Date | CustomerName | State | City | Location |
|---|---|---|---|---|---|
| B-25627 | Monday, April 23, 2018 | Hitika | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25637 | Thursday, April 26, 2018 | Ashmi | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25645 | Tuesday, May 1, 2018 | Yaanvi | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25651 | Monday, May 7, 2018 | Anurag | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25655 | Friday, May 11, 2018 | Nida | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25663 | Saturday, May 19, 2018 | Pournamasi | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25680 | Monday, June 4, 2018 | Aayushi | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25681 | Monday, June 4, 2018 | Bhawna | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25685 | Sunday, June 10, 2018 | Sheetal | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25688 | Monday, June 11, 2018 | Swetha | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25690 | Friday, June 15, 2018 | Gunjan | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25692 | Sunday, June 17, 2018 | Rashmi | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25701 | Tuesday, June 26, 2018 | Maithilee | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25703 | Thursday, June 28, 2018 | Ekta | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25705 | Saturday, June 30, 2018 | Shweta | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25708 | Sunday, July 1, 2018 | Kishwar | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25709 | Sunday, July 1, 2018 | Aakanksha | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25719 | Thursday, July 12, 2018 | Rashmi | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25724 | Thursday, July 19, 2018 | Sheetal | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25727 | Sunday, July 22, 2018 | Turumella | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25734 | Sunday, July 29, 2018 | Pranav | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25737 | Wednesday, August 1, 2018 | Shubham | Madhya Pradesh | Indore | Indore,Madhya Pradesh |
| B-25741 | Friday, August 3, 2018 | Navdeep | Madhya Pradesh | Indore | Indore,Madhya Pradesh |

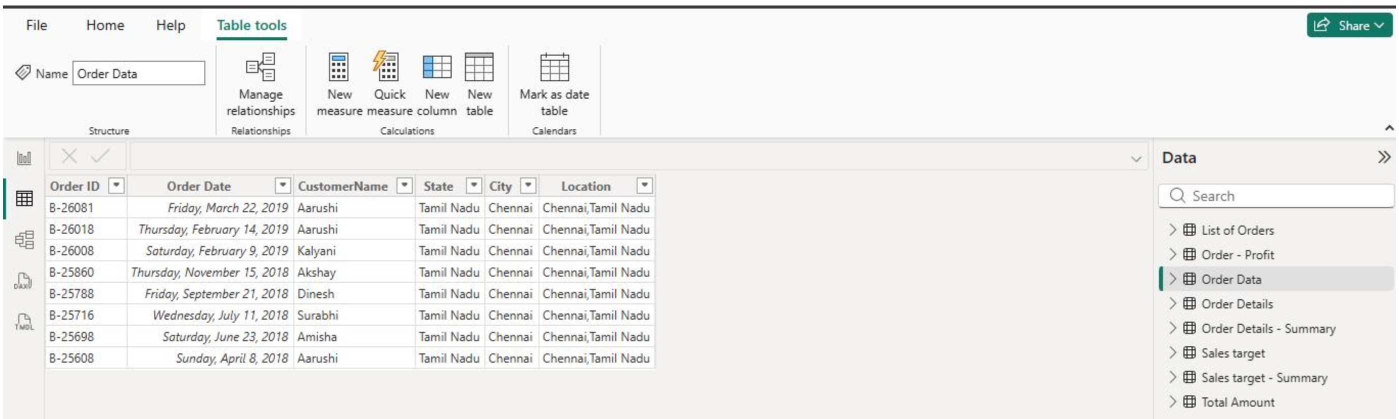
Table: List of Orders (500 rows)

## 2. ORDER - PROFIT



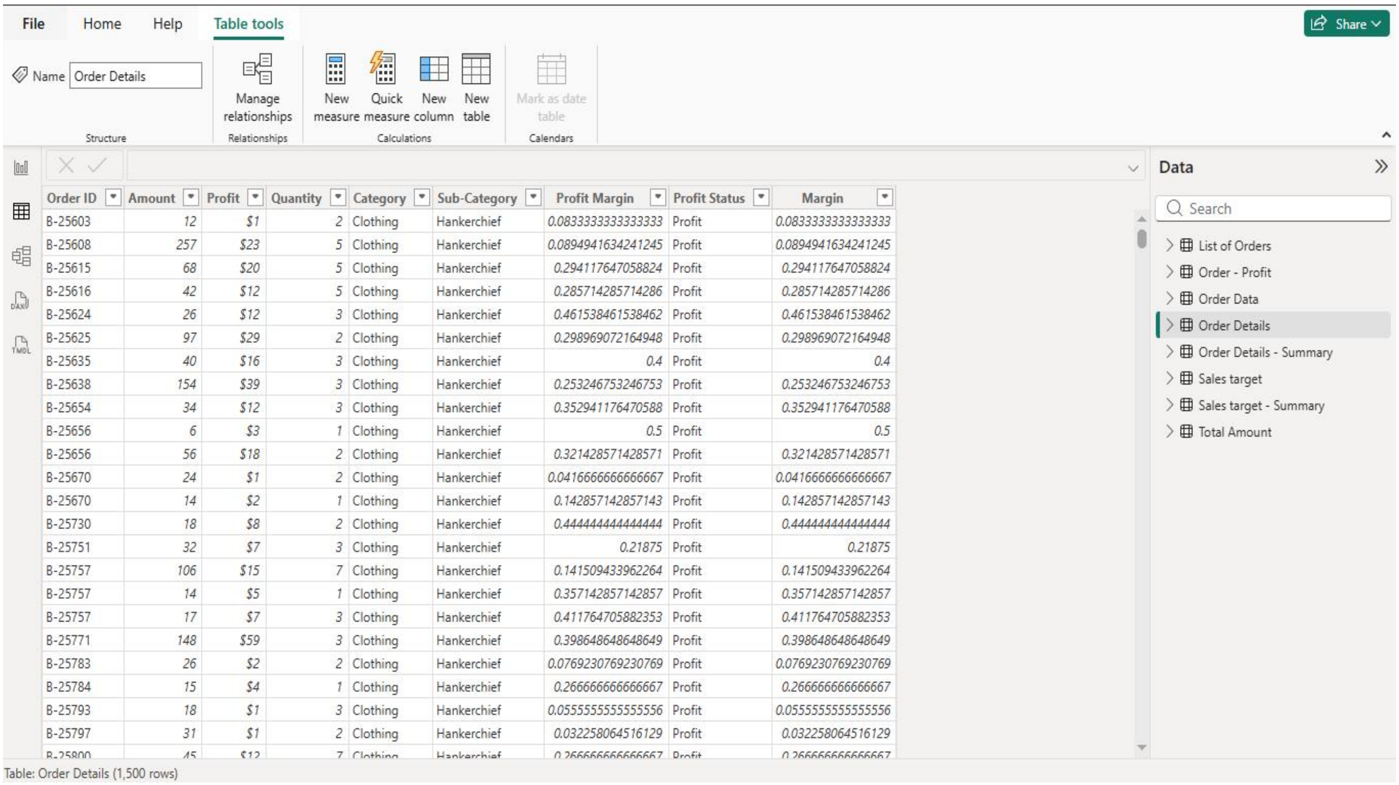
| Category | Average Profit |
|---|---|
| Furniture | 9.45679012345681 |
| Clothing | 11.7629083245522 |
| Electronics | 34.0714285714286 |

## 3. ORDER DATA



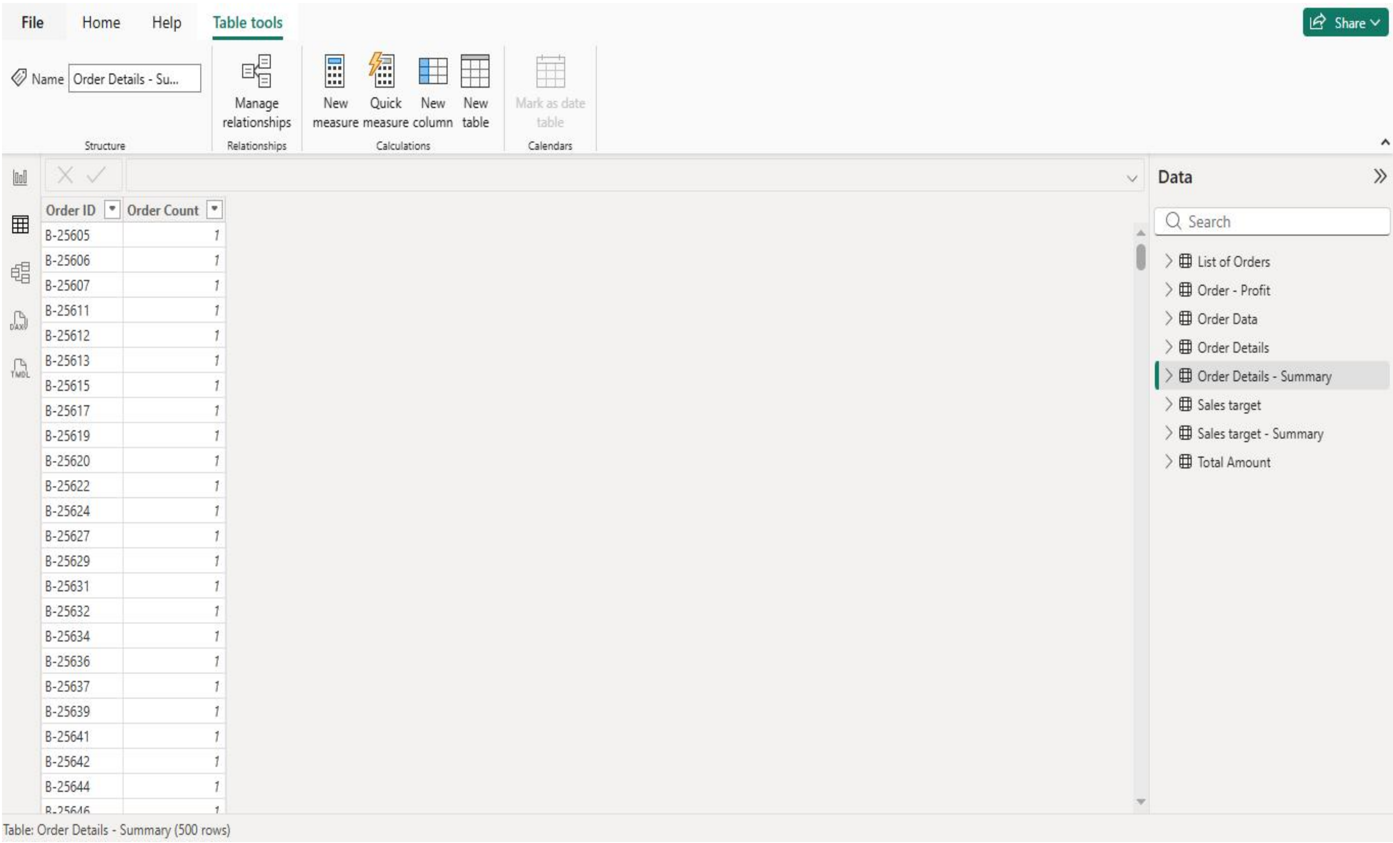
| Order ID | Order Date | CustomerName | State | City | Location |
|---|---|---|---|---|---|
| B-26081 | Friday, March 22, 2019 | Aarushi | Tamil Nadu | Chennai | Chennai,Tamil Nadu |
| B-26018 | Thursday, February 14, 2019 | Aarushi | Tamil Nadu | Chennai | Chennai,Tamil Nadu |
| B-26008 | Saturday, February 9, 2019 | Kalyani | Tamil Nadu | Chennai | Chennai,Tamil Nadu |
| B-25860 | Thursday, November 15, 2018 | Akshay | Tamil Nadu | Chennai | Chennai,Tamil Nadu |
| B-25788 | Friday, September 21, 2018 | Dinesh | Tamil Nadu | Chennai | Chennai,Tamil Nadu |
| B-25716 | Wednesday, July 11, 2018 | Surabhi | Tamil Nadu | Chennai | Chennai,Tamil Nadu |
| B-25698 | Saturday, June 23, 2018 | Amisha | Tamil Nadu | Chennai | Chennai,Tamil Nadu |
| B-25608 | Sunday, April 8, 2018 | Aarushi | Tamil Nadu | Chennai | Chennai,Tamil Nadu |

## 4. ORDER  DETAILS



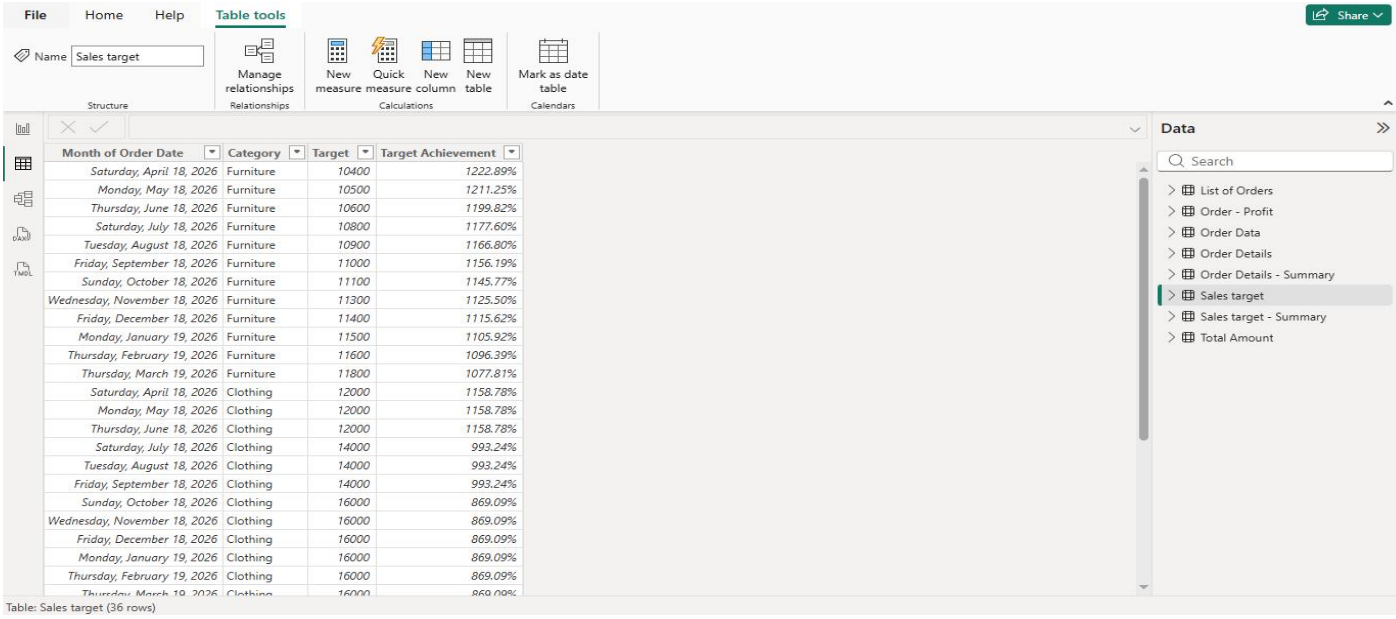

| Order ID | Amount | Profit | Quantity | Category | Sub-Category | Profit Margin | Profit Status | Margin |
|---|---|---|---|---|---|---|---|---|
| B-25603 | 12 | $1 | 2 | Clothing | Hankerchief | 0.0833333333333333 | Profit | 0.0833333333333333 |
| B-25608 | 257 | $23 | 5 | Clothing | Hankerchief | 0.0894941634241245 | Profit | 0.0894941634241245 |
| B-25615 | 68 | $20 | 5 | Clothing | Hankerchief | 0.294117647058824 | Profit | 0.294117647058824 |
| B-25616 | 42 | $12 | 5 | Clothing | Hankerchief | 0.285714285714286 | Profit | 0.285714285714286 |
| B-25624 | 26 | $12 | 3 | Clothing | Hankerchief | 0.461538461538462 | Profit | 0.461538461538462 |
| B-25625 | 97 | $29 | 2 | Clothing | Hankerchief | 0.298969072164948 | Profit | 0.298969072164948 |
| B-25635 | 40 | $16 | 3 | Clothing | Hankerchief | 0.4 | Profit | 0.4 |
| B-25638 | 154 | $39 | 3 | Clothing | Hankerchief | 0.253246753246753 | Profit | 0.253246753246753 |
| B-25654 | 34 | $12 | 3 | Clothing | Hankerchief | 0.352941176470588 | Profit | 0.352941176470588 |
| B-25656 | 6 | $3 | 1 | Clothing | Hankerchief | 0.5 | Profit | 0.5 |
| B-25656 | 56 | $18 | 2 | Clothing | Hankerchief | 0.321428571428571 | Profit | 0.321428571428571 |
| B-25670 | 24 | $1 | 2 | Clothing | Hankerchief | 0.0416666666666667 | Profit | 0.0416666666666667 |
| B-25670 | 14 | $2 | 1 | Clothing | Hankerchief | 0.142857142857143 | Profit | 0.142857142857143 |
| B-25730 | 18 | $8 | 2 | Clothing | Hankerchief | 0.444444444444444 | Profit | 0.444444444444444 |
| B-25751 | 32 | $7 | 3 | Clothing | Hankerchief | 0.21875 | Profit | 0.21875 |
| B-25757 | 106 | $15 | 7 | Clothing | Hankerchief | 0.141509433962264 | Profit | 0.141509433962264 |
| B-25757 | 14 | $5 | 1 | Clothing | Hankerchief | 0.357142857142857 | Profit | 0.357142857142857 |
| B-25757 | 17 | $7 | 3 | Clothing | Hankerchief | 0.411764705882353 | Profit | 0.411764705882353 |
| B-25771 | 148 | $59 | 3 | Clothing | Hankerchief | 0.398648648648649 | Profit | 0.398648648648649 |
| B-25783 | 26 | $2 | 2 | Clothing | Hankerchief | 0.0769230769230769 | Profit | 0.0769230769230769 |
| B-25784 | 15 | $4 | 1 | Clothing | Hankerchief | 0.266666666666667 | Profit | 0.266666666666667 |
| B-25793 | 18 | $1 | 3 | Clothing | Hankerchief | 0.0555555555555556 | Profit | 0.0555555555555556 |
| B-25797 | 31 | $1 | 2 | Clothing | Hankerchief | 0.032258064516129 | Profit | 0.032258064516129 |
| B-25800 | 45 | $12 | 7 | Clothing | Hankerchief | 0.266666666666667 | Profit | 0.266666666666667 |

# 5. ORDER DETAILS - SUMMARY



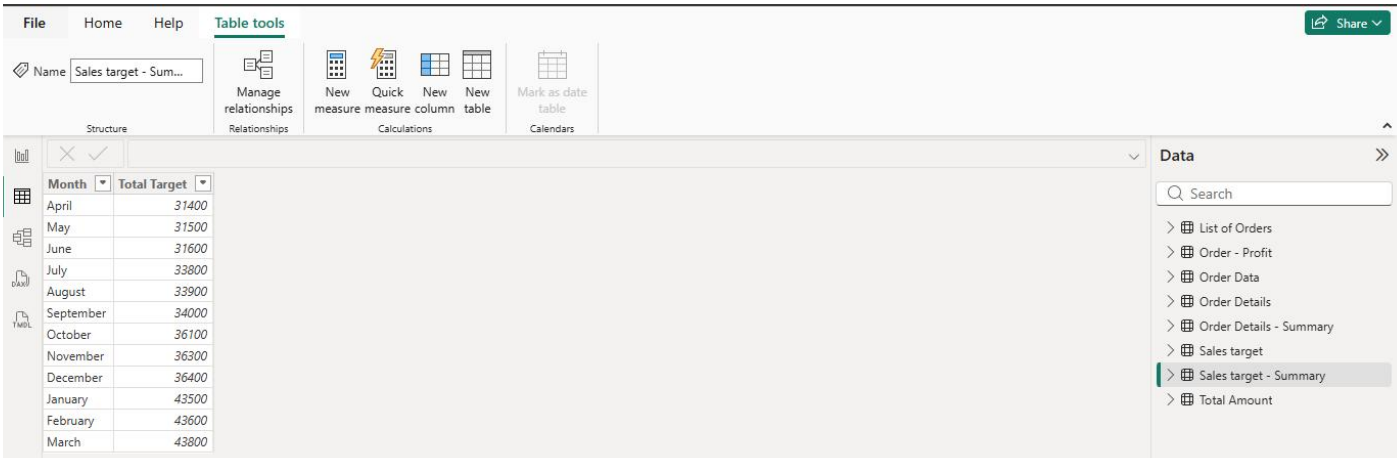

| Order ID | Order Count |
|---|---|
| B-25605 | 1 |
| B-25606 | 1 |
| B-25607 | 1 |
| B-25611 | 1 |
| B-25612 | 1 |
| B-25613 | 1 |
| B-25615 | 1 |
| B-25617 | 1 |
| B-25619 | 1 |
| B-25620 | 1 |
| B-25622 | 1 |
| B-25624 | 1 |
| B-25627 | 1 |
| B-25629 | 1 |
| B-25631 | 1 |
| B-25632 | 1 |
| B-25634 | 1 |
| B-25636 | 1 |
| B-25637 | 1 |
| B-25639 | 1 |
| B-25641 | 1 |
| B-25642 | 1 |
| B-25644 | 1 |
| B-25646 | 1 |

# 6. SALES TARGET



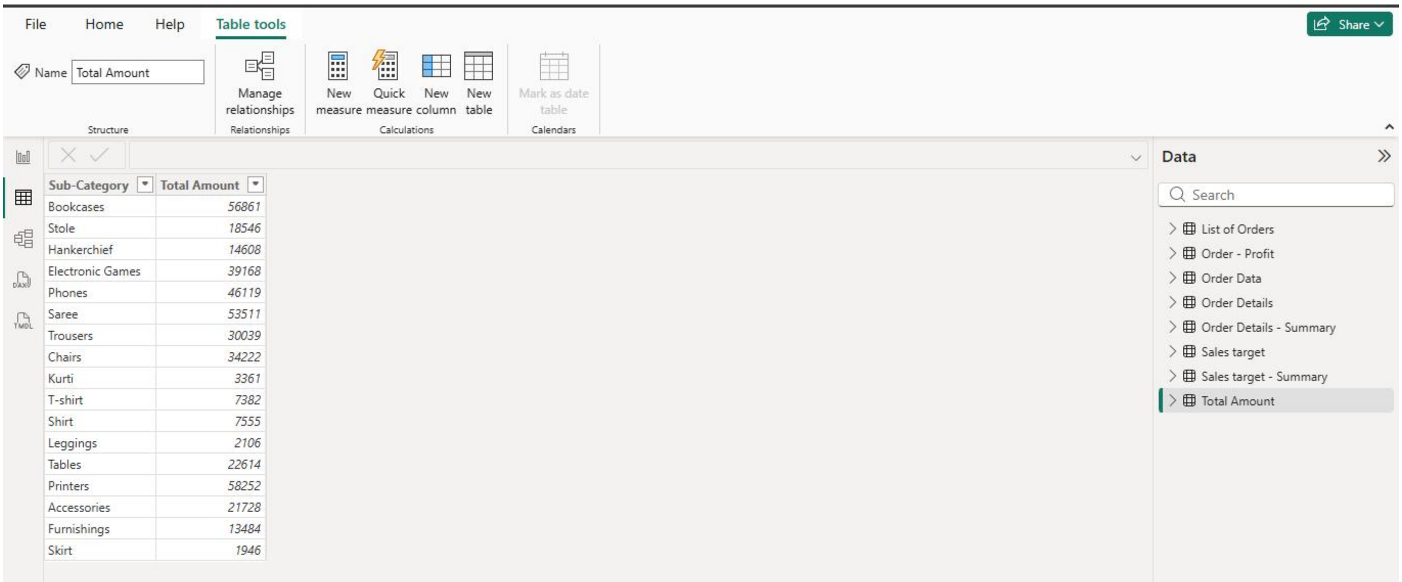

| Month of Order Date | Category | Target | Target Achievement |
|---|---|---|---|
| Saturday, April 18, 2026 | Furniture | 10400 | 1222.89% |
| Monday, May 18, 2026 | Furniture | 10500 | 1211.25% |
| Thursday, June 18, 2026 | Furniture | 10600 | 1199.82% |
| Saturday, July 18, 2026 | Furniture | 10800 | 1177.60% |
| Tuesday, August 18, 2026 | Furniture | 10900 | 1166.80% |
| Friday, September 18, 2026 | Furniture | 11000 | 1156.19% |
| Sunday, October 18, 2026 | Furniture | 11100 | 1145.77% |
| Wednesday, November 18, 2026 | Furniture | 11300 | 1125.50% |
| Friday, December 18, 2026 | Furniture | 11400 | 1115.62% |
| Monday, January 19, 2026 | Furniture | 11500 | 1105.92% |
| Thursday, February 19, 2026 | Furniture | 11600 | 1096.39% |
| Thursday, March 19, 2026 | Furniture | 11800 | 1077.81% |
| Saturday, April 18, 2026 | Clothing | 12000 | 1158.78% |
| Monday, May 18, 2026 | Clothing | 12000 | 1158.78% |
| Thursday, June 18, 2026 | Clothing | 12000 | 1158.78% |
| Saturday, July 18, 2026 | Clothing | 14000 | 993.24% |
| Tuesday, August 18, 2026 | Clothing | 14000 | 993.24% |
| Friday, September 18, 2026 | Clothing | 14000 | 993.24% |
| Sunday, October 18, 2026 | Clothing | 16000 | 869.09% |
| Wednesday, November 18, 2026 | Clothing | 16000 | 869.09% |
| Friday, December 18, 2026 | Clothing | 16000 | 869.09% |
| Monday, January 19, 2026 | Clothing | 16000 | 869.09% |
| Thursday, February 19, 2026 | Clothing | 16000 | 869.09% |
| Thursday, March 19, 2026 | Clothing | 16000 | 869.09% |

# 7. SALES TARGET - SUMMARY




| Month | Total Target |
|---|---|
| April | 31400 |
| May | 31500 |
| June | 31600 |
| July | 33800 |
| August | 33900 |
| September | 34000 |
| October | 36100 |
| November | 36300 |
| December | 36400 |
| January | 43500 |
| February | 43600 |
| March | 43800 |

# 8. TOTAL AMOUNT




| Sub-Category | Total Amount |
|---|---|
| Bookcases | 56861 |
| Stole | 18546 |
| Hankerchief | 14608 |
| Electronic Games | 39168 |
| Phones | 46119 |
| Saree | 53511 |
| Trousers | 30039 |
| Chairs | 34222 |
| Kurti | 3361 |
| T-shirt | 7382 |
| Shirt | 7555 |
| Leggings | 2106 |
| Tables | 22614 |
| Printers | 58252 |
| Accessories | 21728 |
| Furnishings | 13484 |
| Skirt | 1946 |